

CALIBRATING MAGNETIC VARIOMETERS USING QUANTITATIVE ERROR METRICS

Julia Berglind & Peyton Chandler, Department of Geophysics

Advisors: Rich Krahenbuhl (CSM), Brian Shiro (USGS),
Adam Ringler (USGS), Josh Rigler (USGS), & John Spritzer
(USGS)

MOTIVATION & PROBLEM



Bartington Mag690 Variometer



USGS Geomagnetic Observatory

- The USGS has deployed many Bartington Mag690-100s for geomagnetic monitoring
- Optimized for resolving temporal variations, not absolute field values
 - Creates an accuracy problem when they are used in isolation
- Network is expanding into regions without nearby observatories, creating an observational gap

OBJECTIVES & HYPOTHESIS

Quantify the error characteristics of the Mag690-100 relative to a known reference

Develop quantitative metrics (bias, drift rate) benchmarked against the IGRF

Assess whether corrected variometer data can meet accuracy thresholds

*Hypothesis: Systematic errors in Mag690-100 deployments are **quantifiable** via **observatory comparison** and **correctable** using **IGRF-referenced metrics**, enabling reliable **absolute field estimation** at sites without co-located observatories.*



METHODS & APPROACH

PHASE I: BOU DEPLOYMENT

Part One



1/30/2026 – 2/06/2026

Wire + solder variometer
in USGS geomagnetic lab

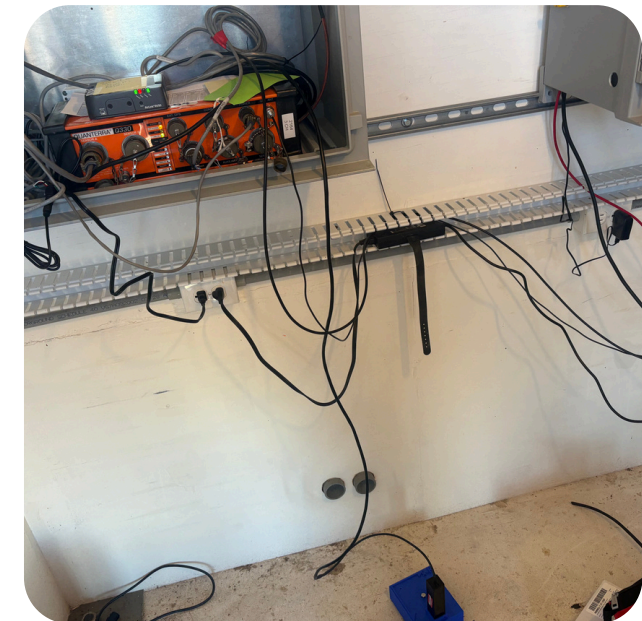
Part Two



2/20/2026

Get GPS points for test
cube location

Part Three



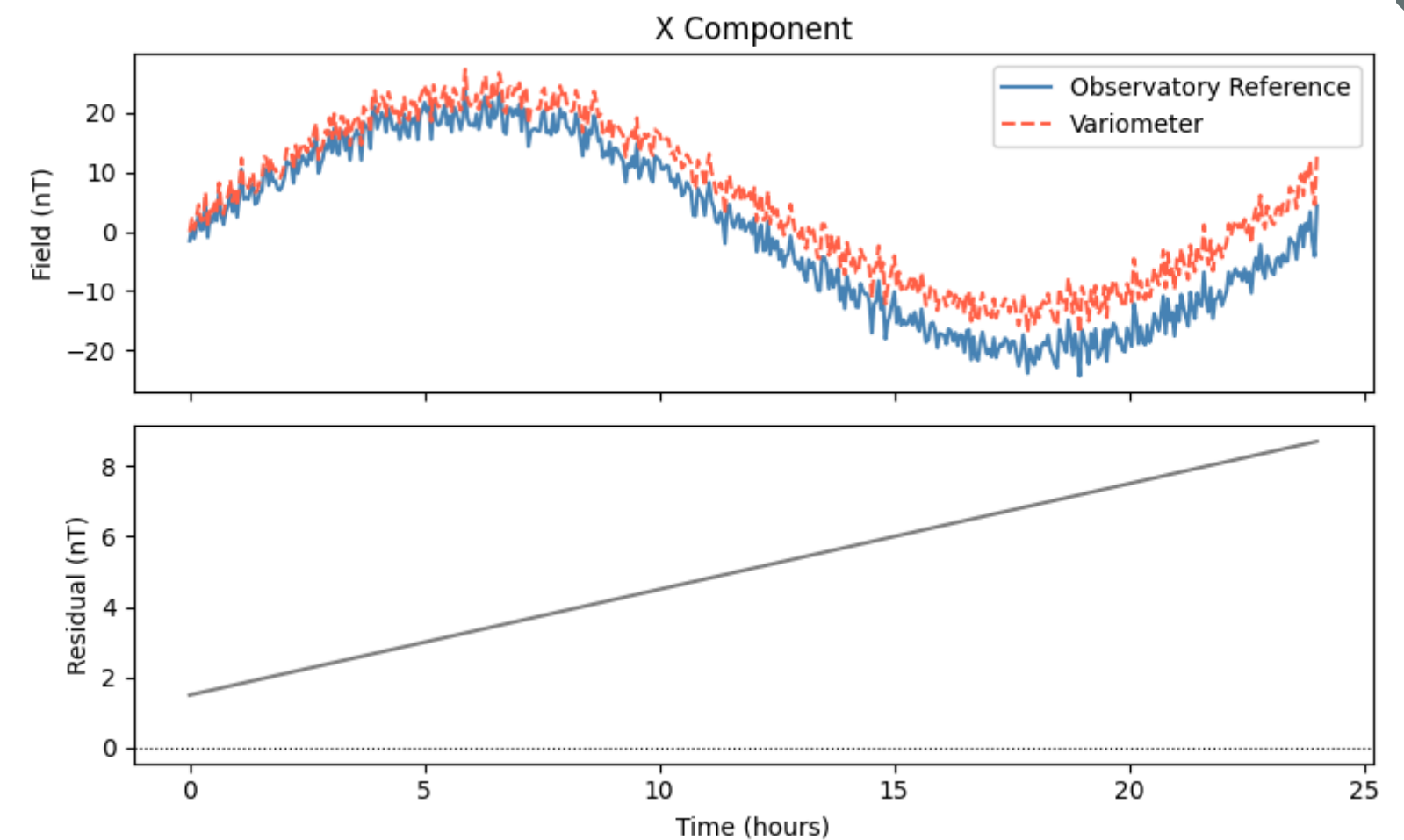
2/27/2026

Install variometer

PHASE 2: ERROR METRIC DEVELOPMENT

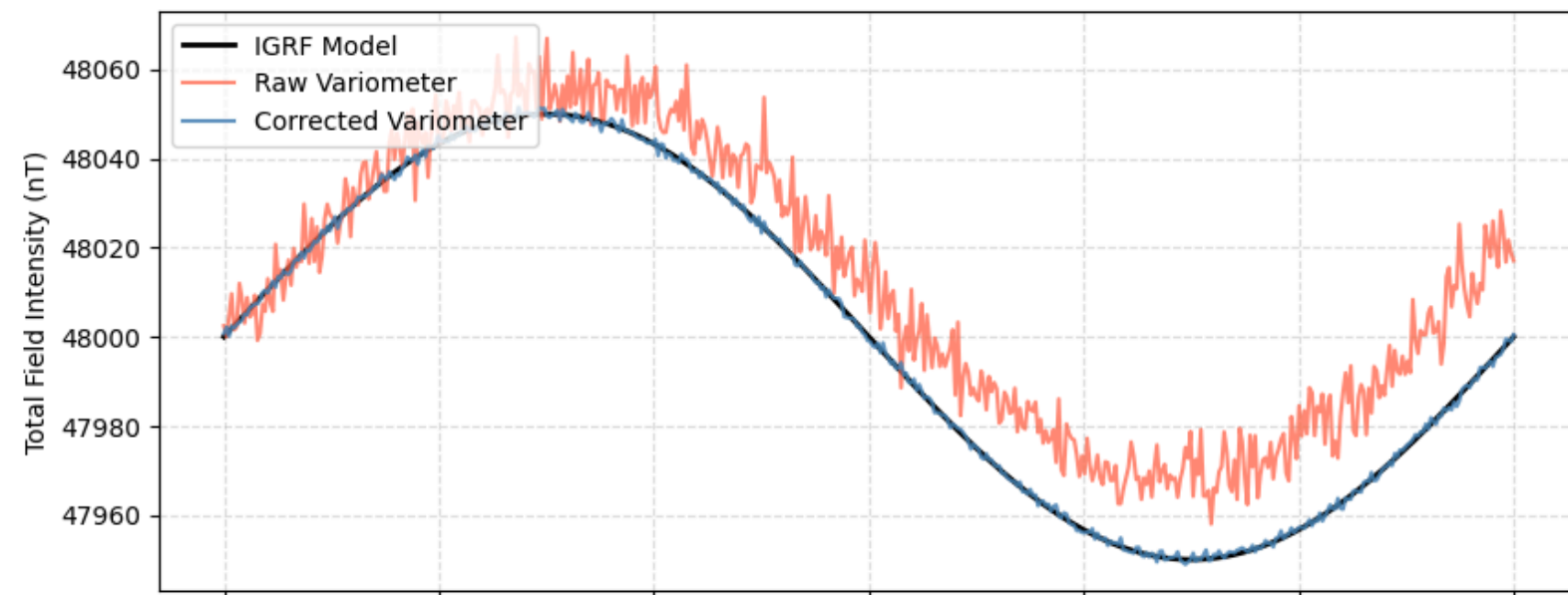
Goal: Compute error metric between variometer metrics and observatory data

- Candidate metrics:
 - Baseline offset per component
 - Secular drift rate
 - IGRF residuals
 - Sensitivity coefficients
 - Angular misalignment estimation



Schematic of X-component residual

PHASE 3: VALIDATION



Schematic of corrected variometer data with respect to IGRF

- Apply correction scheme to independent variometer stations
- Success criterion: residuals within $\pm 5\text{nT}$ (INTERMAGNET standard)
- Boulder observatory provides ground-truth reference

SIGNIFICANCE & NEXT STEPS

Why this matters:

- Provides USGS a validated correction protocol to extract absolute field values from existing network deployments
- Improves data quality for space weather monitoring, geomagnetic storm detection, and long-wavelength field modeling

Next Steps:

- View initial data from Boulder
- Begin Observatory comparison
- Develop error metric pipeline
- Apply metrics to independent USGS stations



QUESTIONS